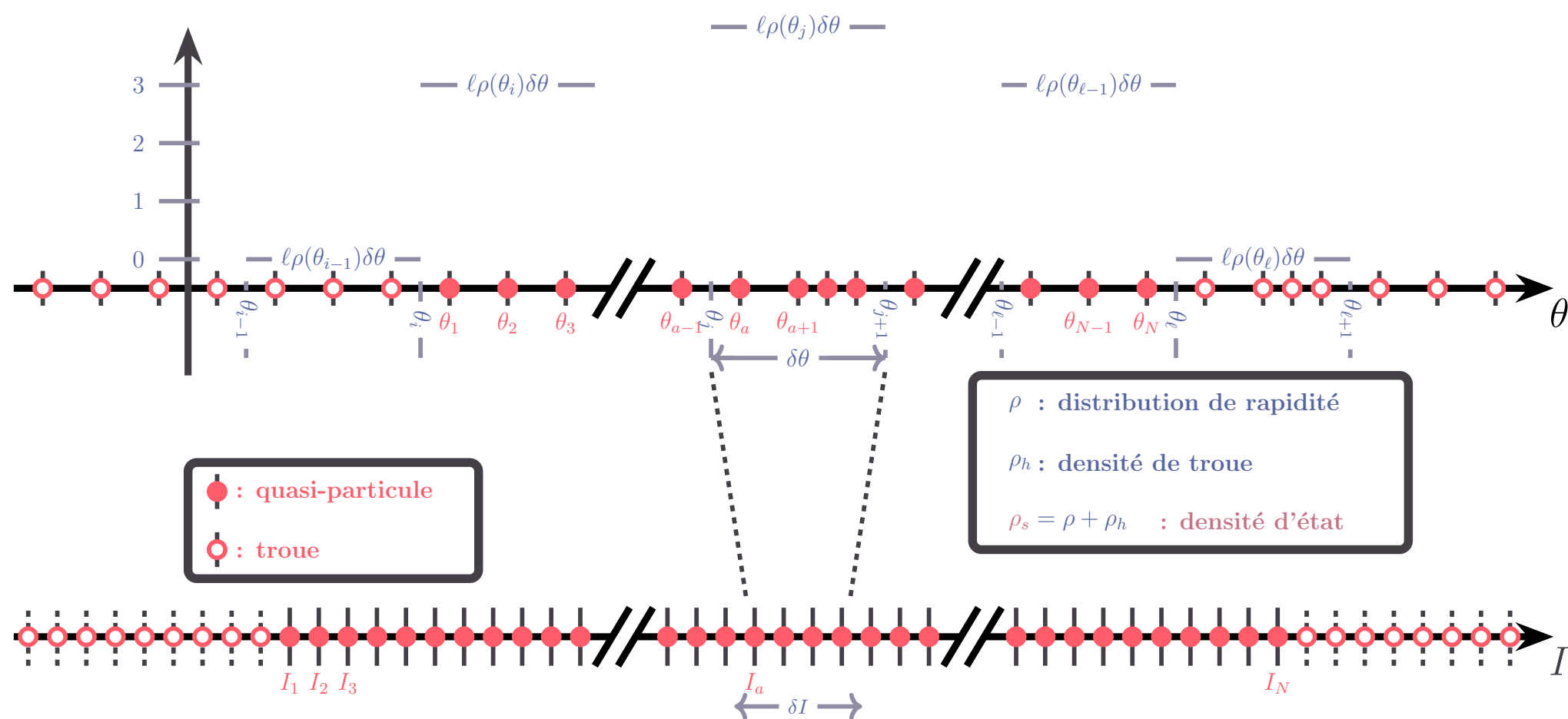
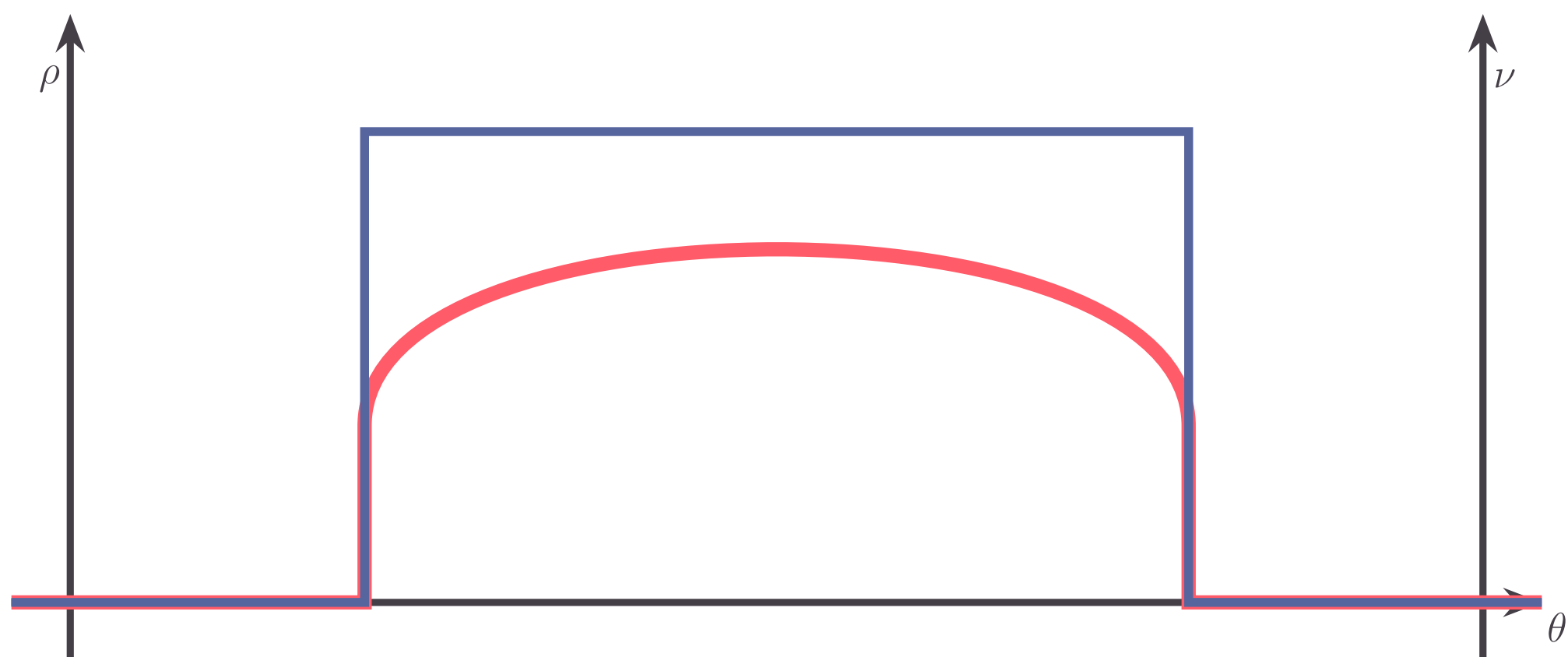
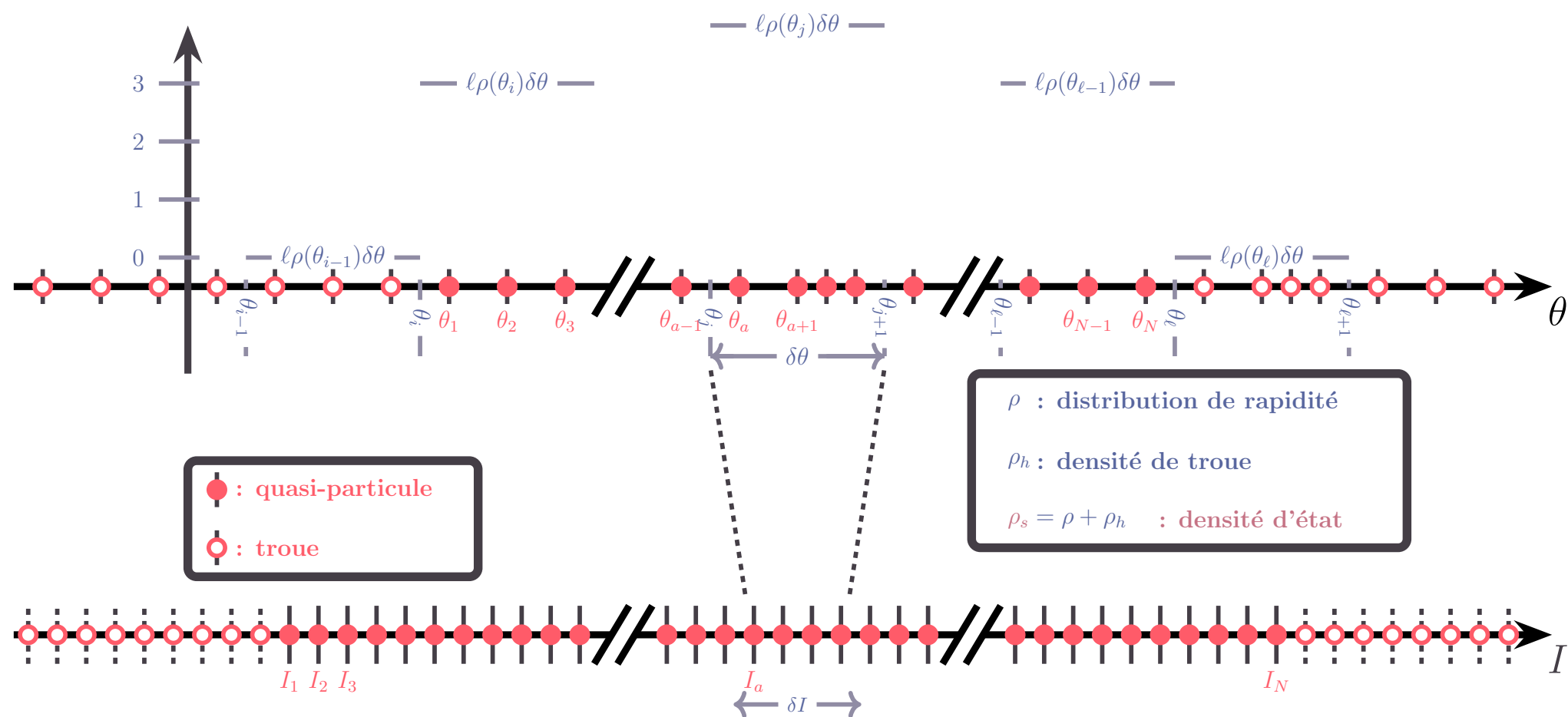
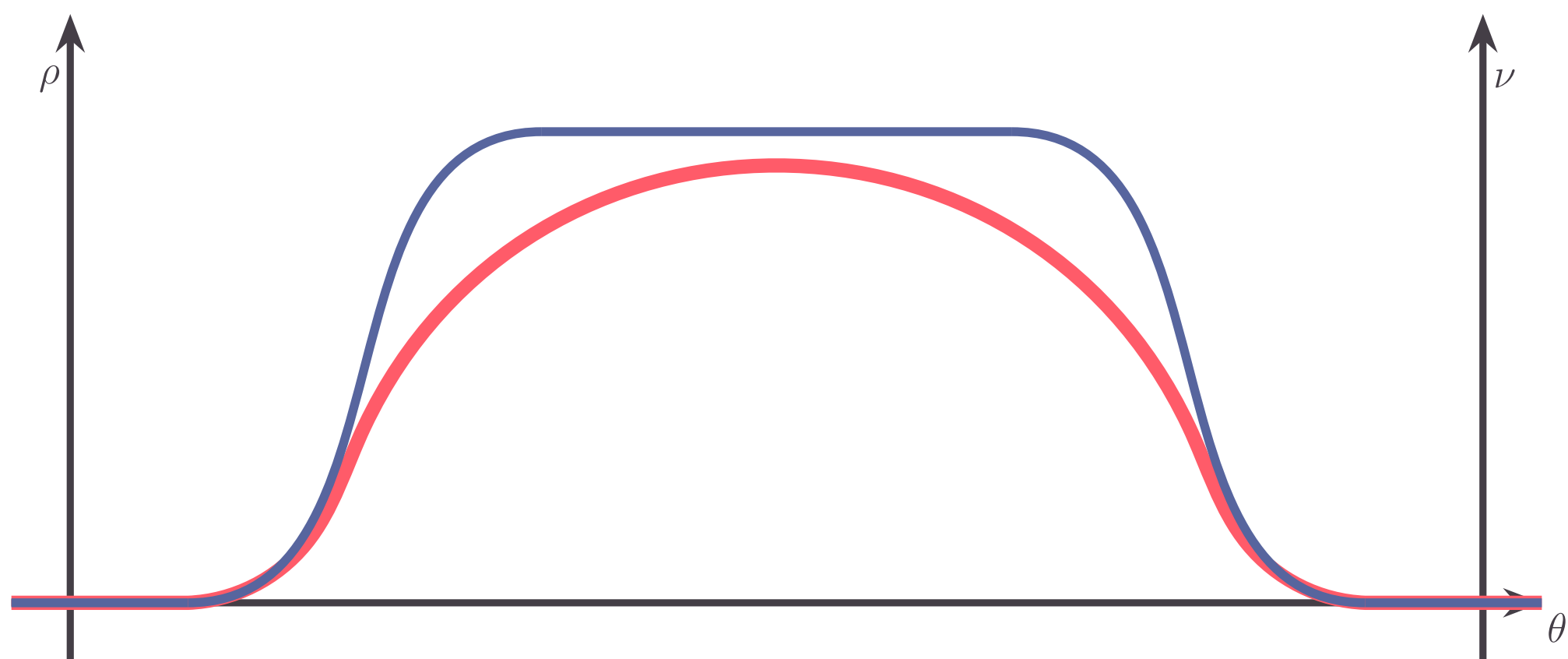
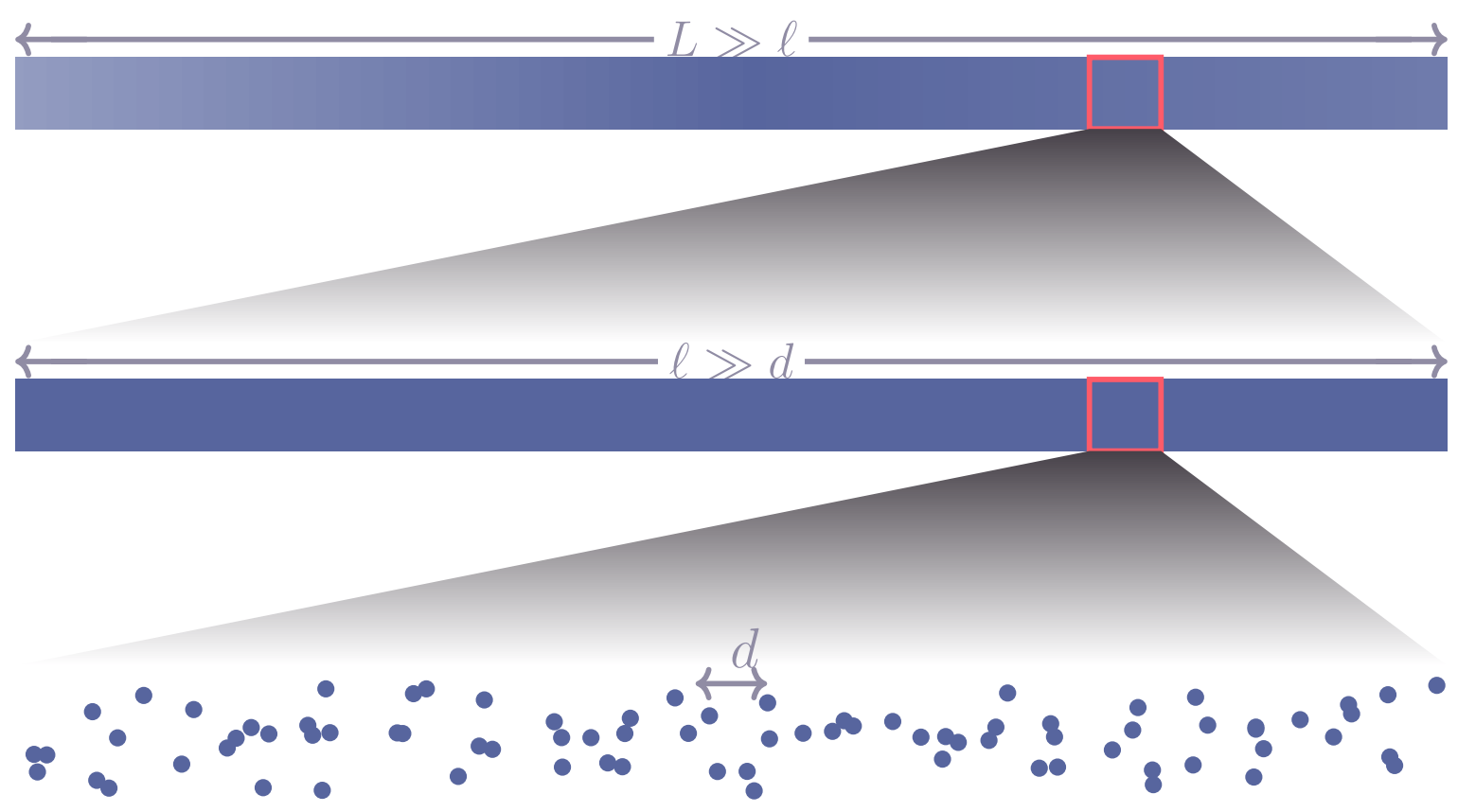
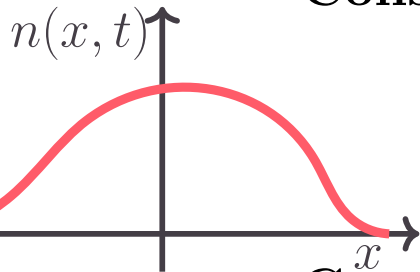






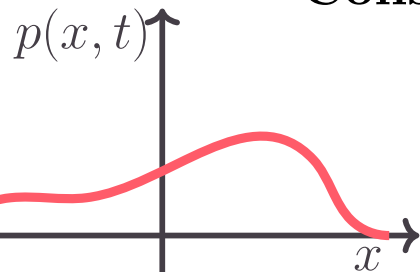


Conservation of the atom number :



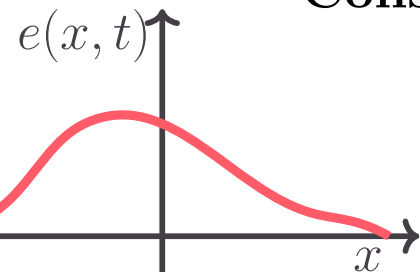
$$\int dx n(x, t) = N$$
$$\partial_t n + \partial_x j_n = 0$$

Conservation of total momentum:

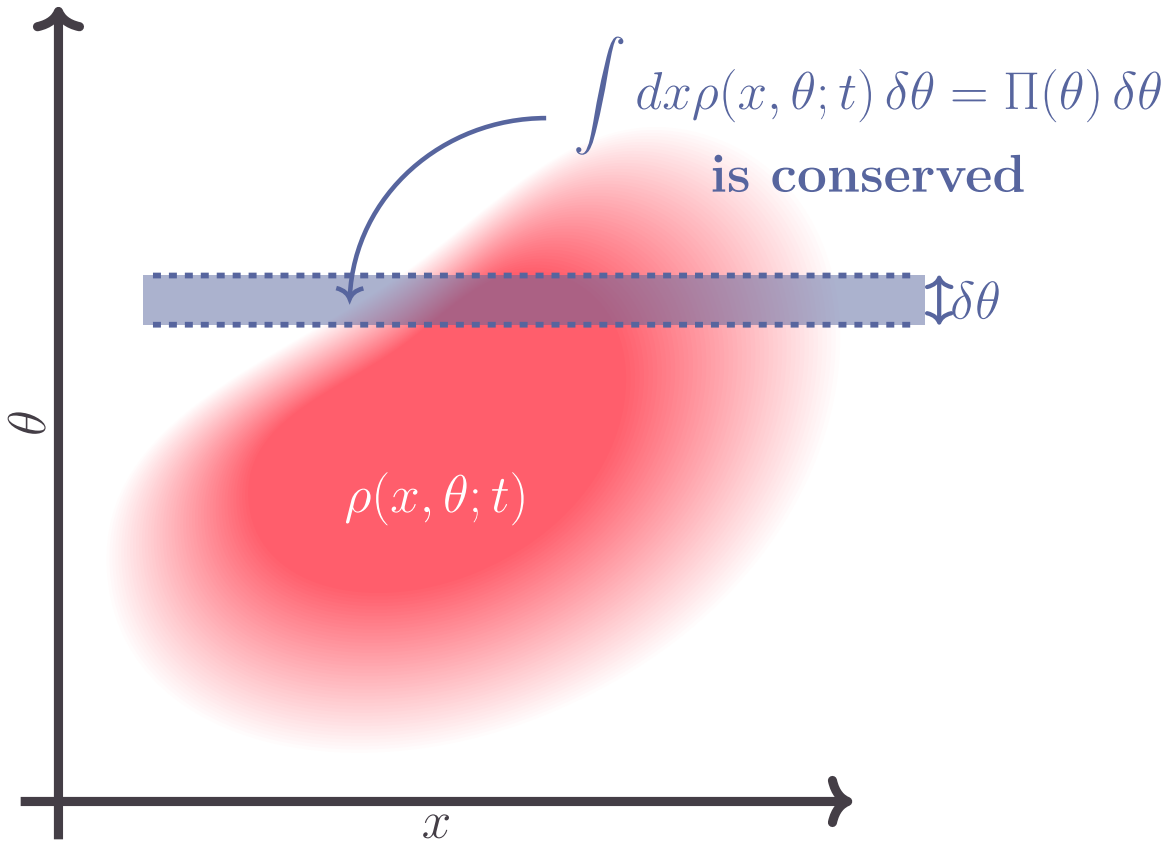


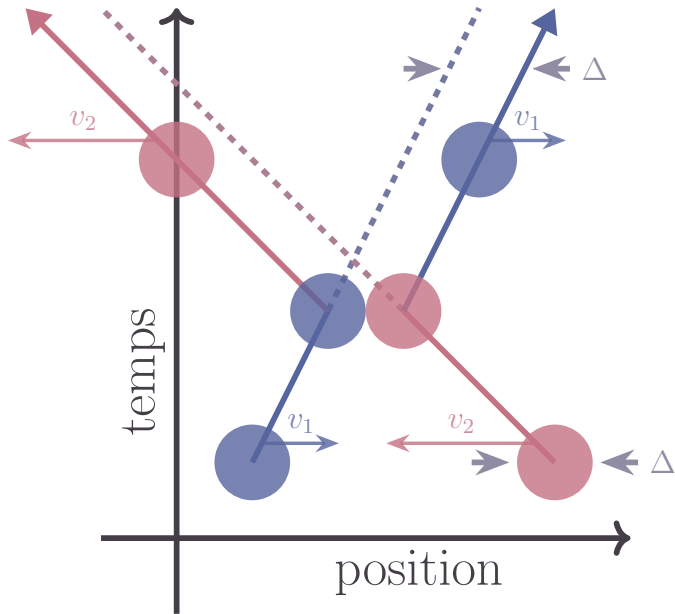
$$\int dx p(x, t) = P$$
$$\partial_t p + \partial_x j_p = 0$$

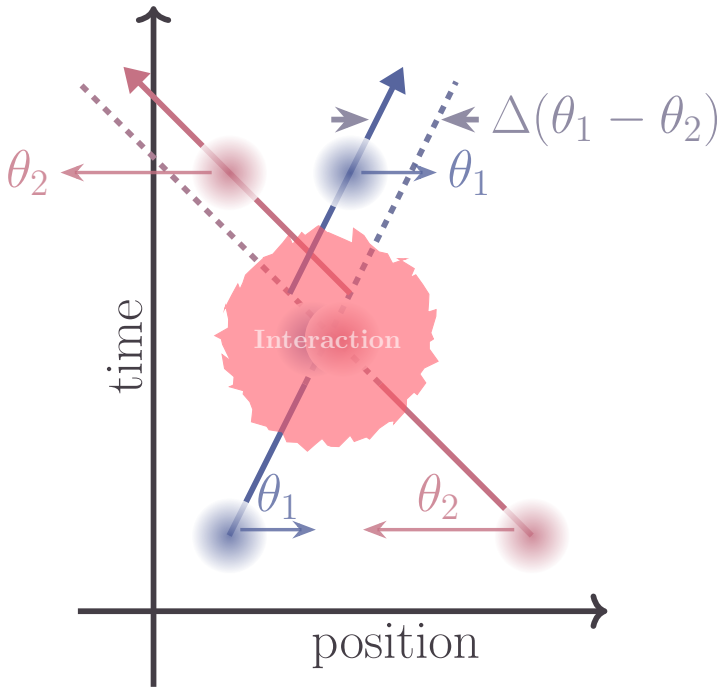
Conservation of total energy :



$$\int dx e(x, t) = E$$
$$\partial_t e + \partial_x j_e = 0$$







- diameters of balls

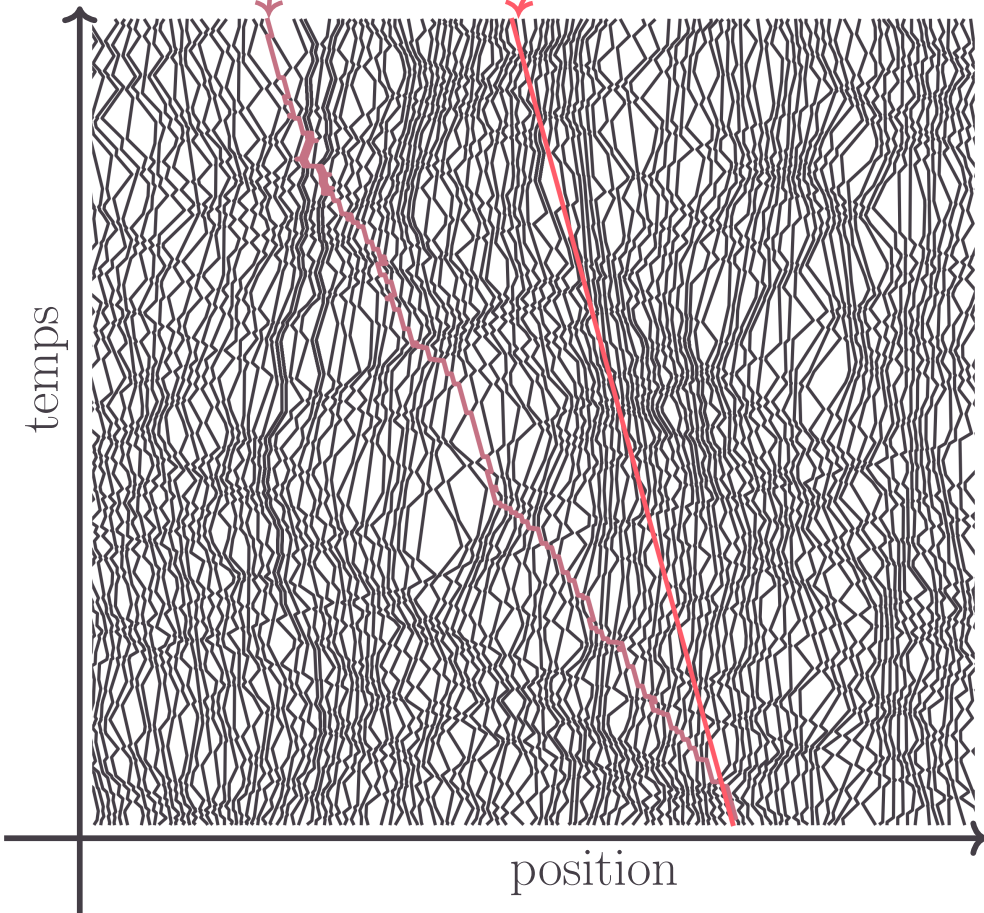
$$v_{[\rho]}^{\text{eff}}(v) = v + \Delta \int (v_{[\rho]}^{\text{eff}}(w) - v_{[\rho]}^{\text{eff}}(v)) \rho(w) dw$$

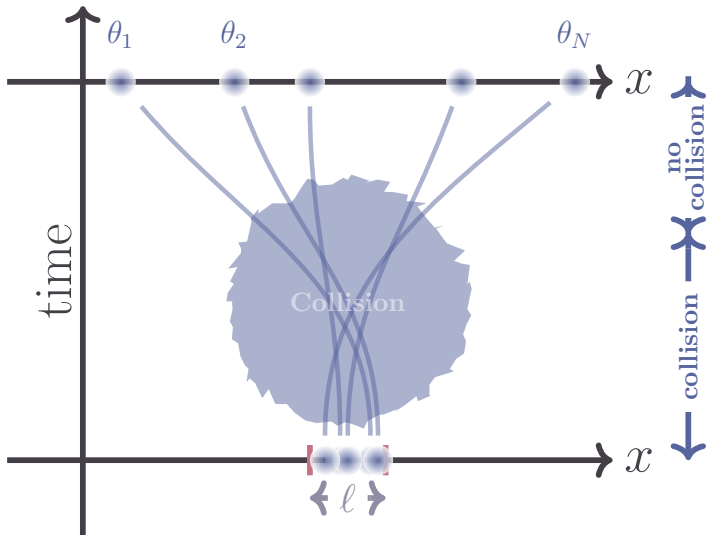
number of collisions per second

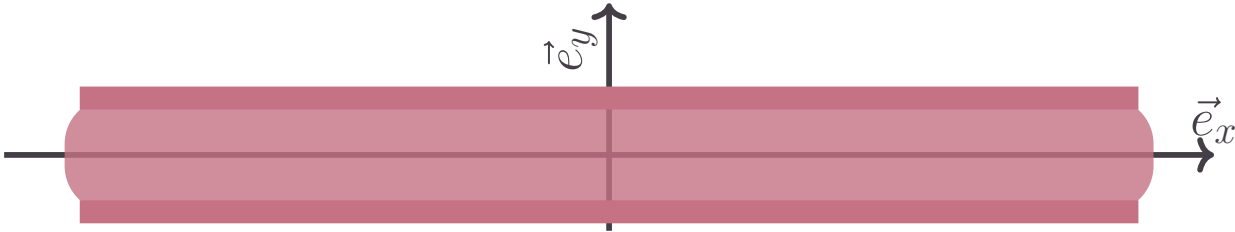
v

temps

position

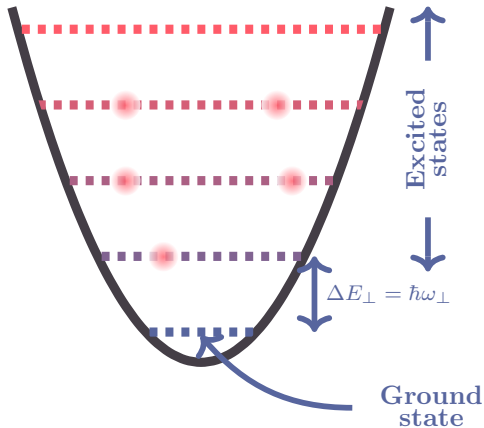
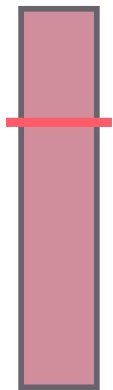






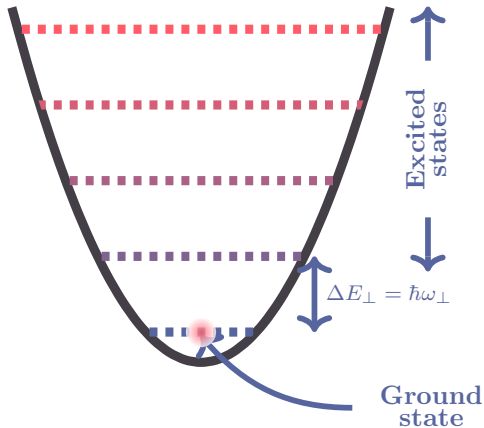
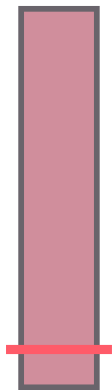
Energy/Atom

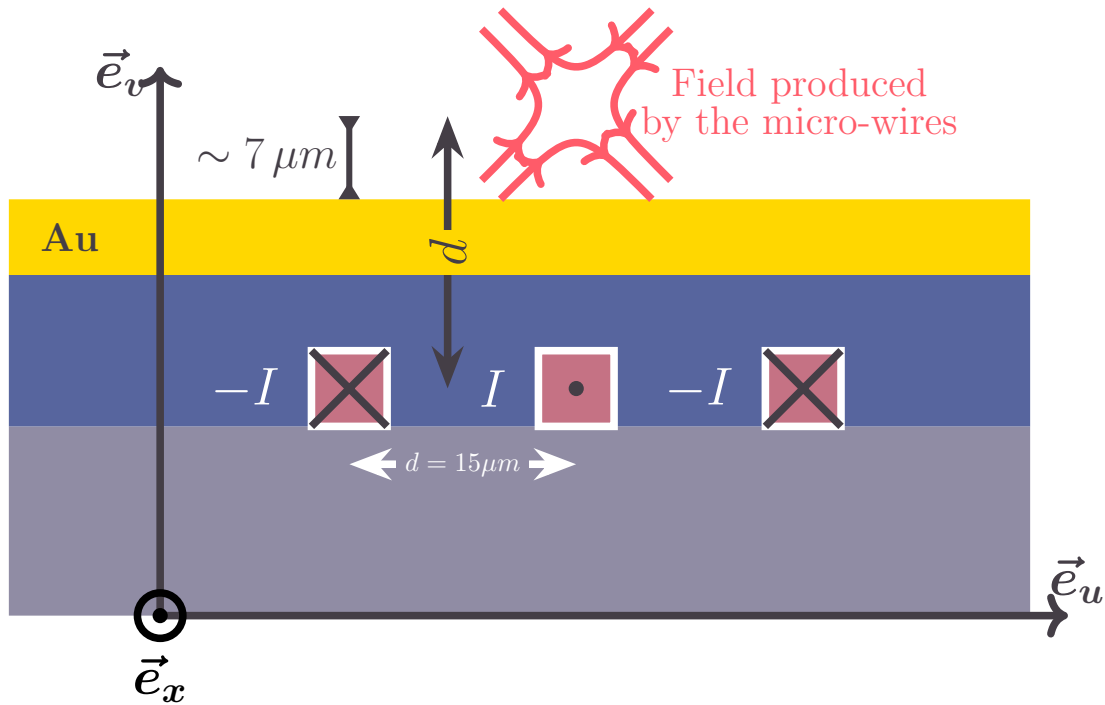
Transverse energy states

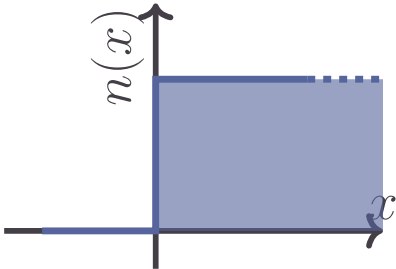


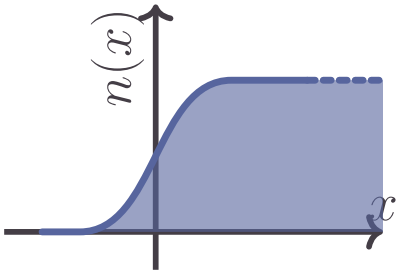
Energy/Atom

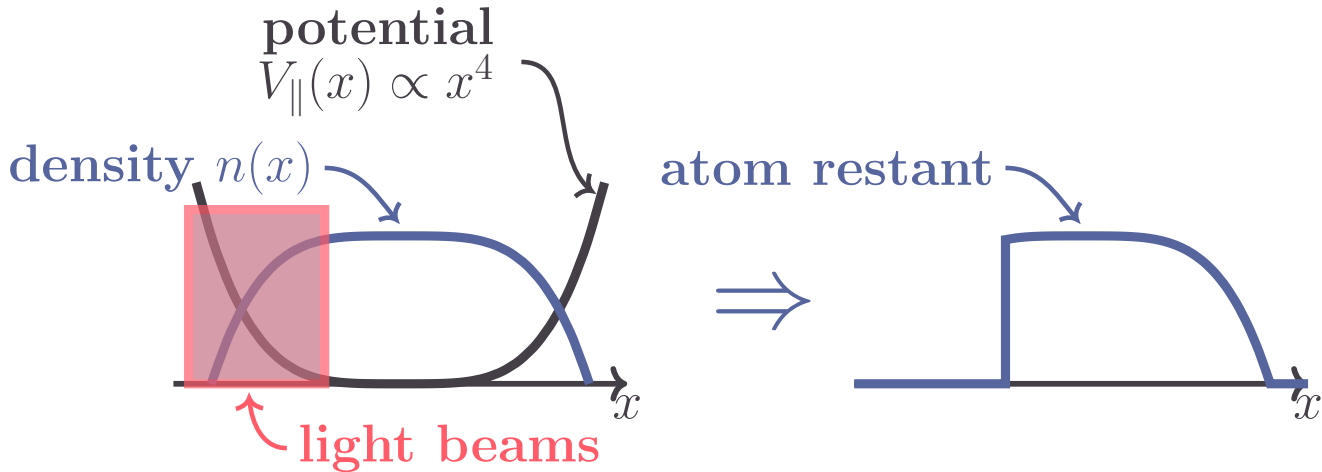
Transverse energy states

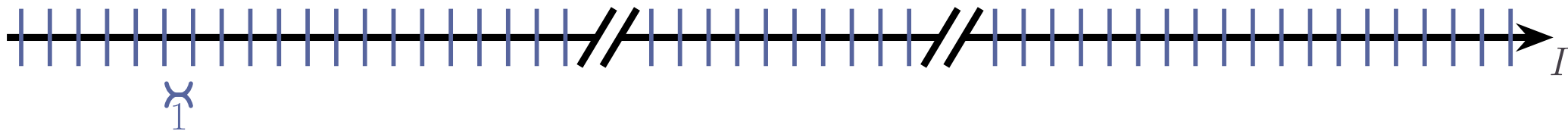


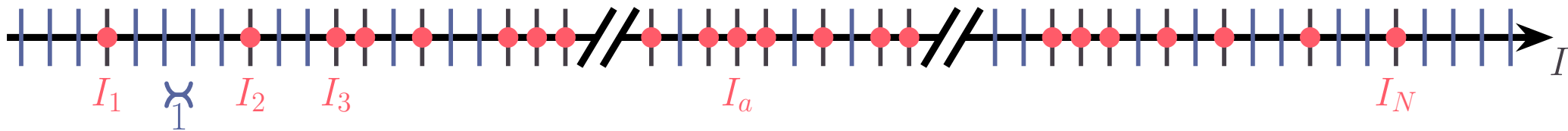


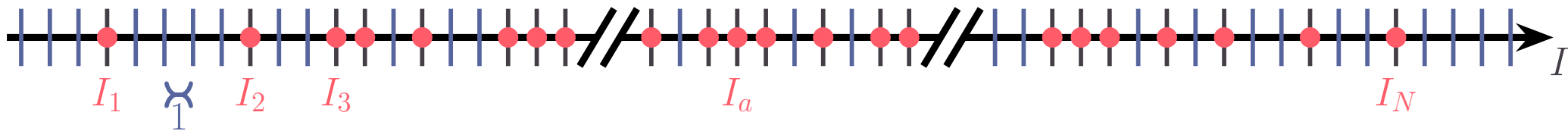


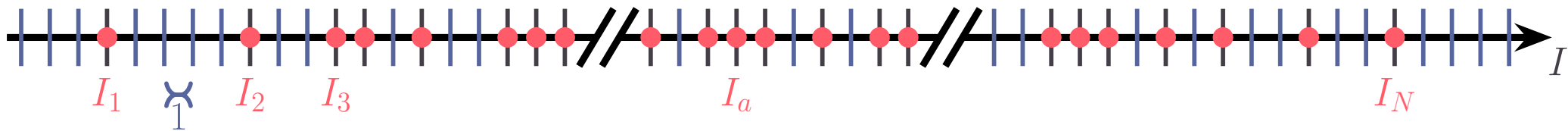
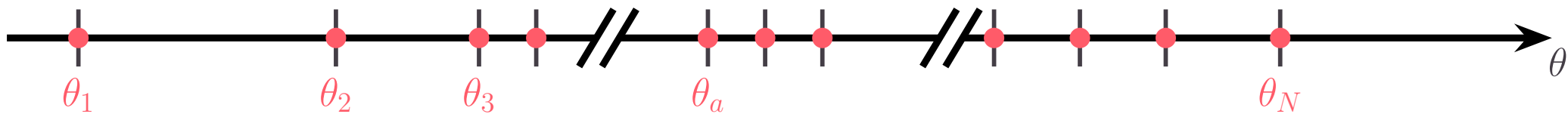


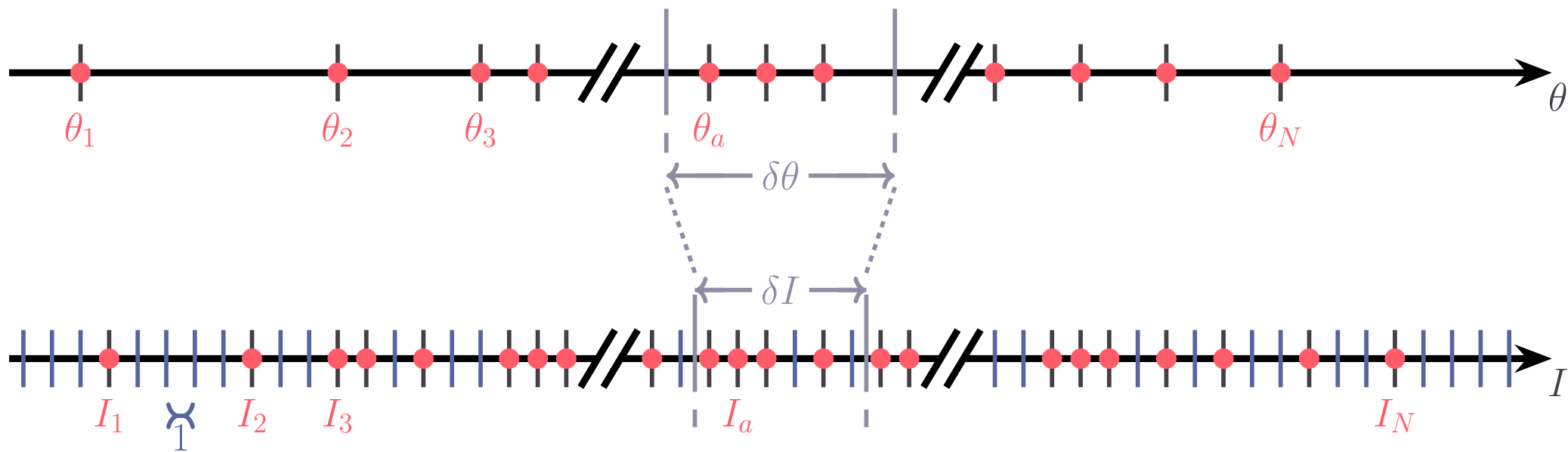


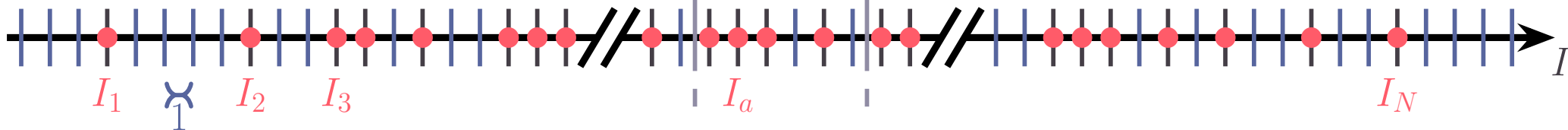
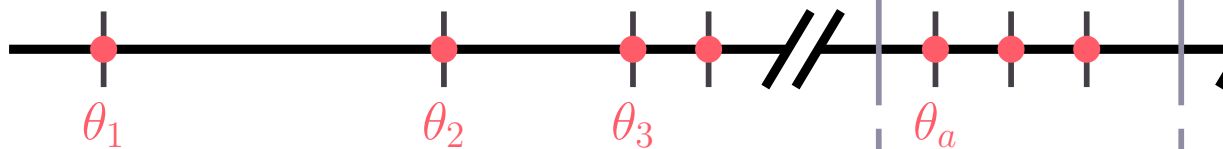
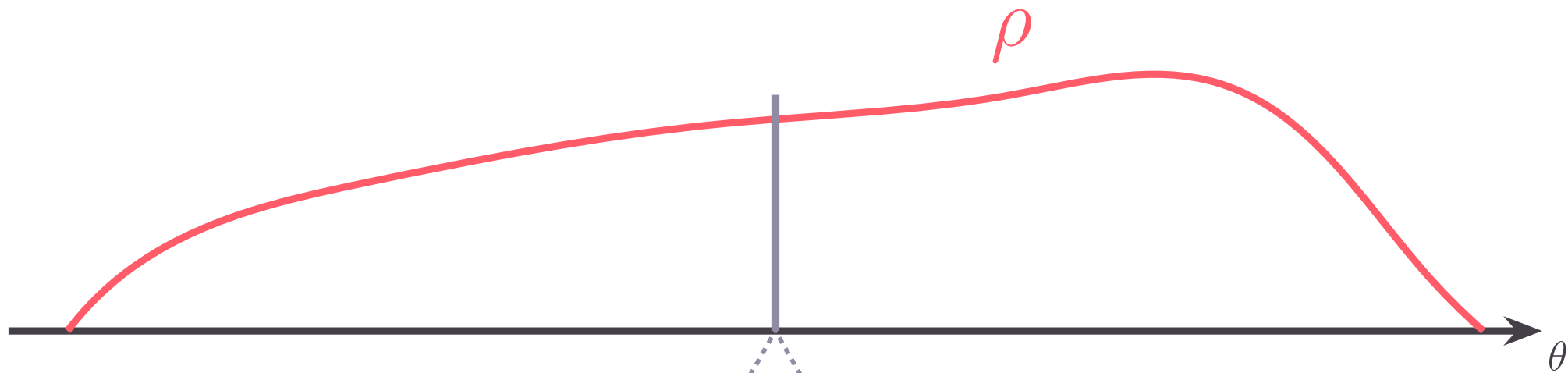










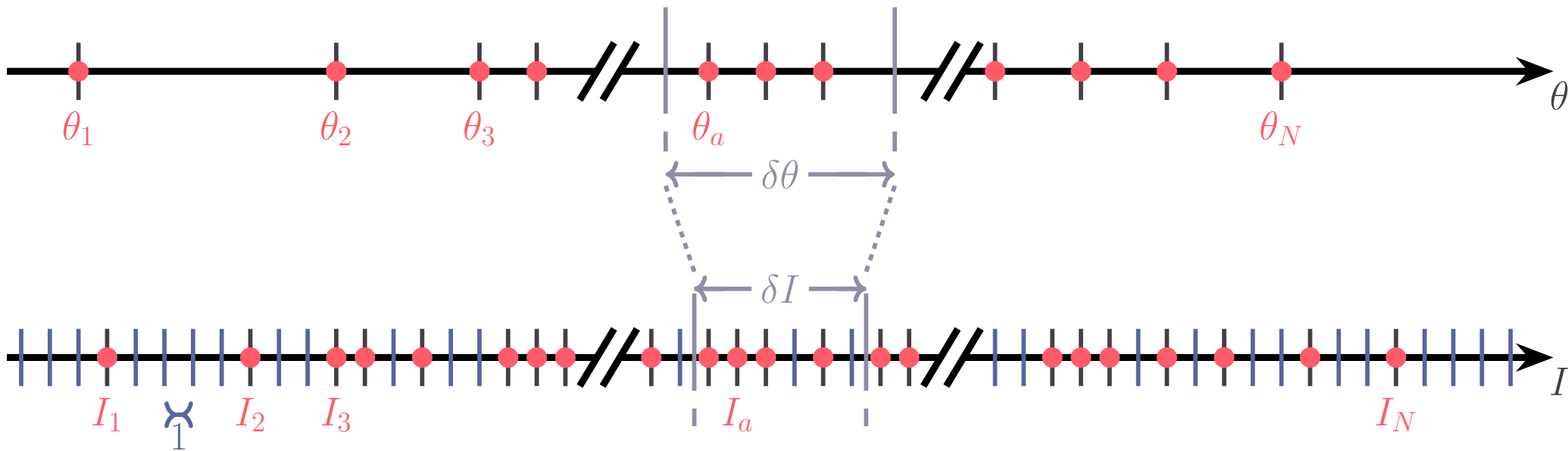
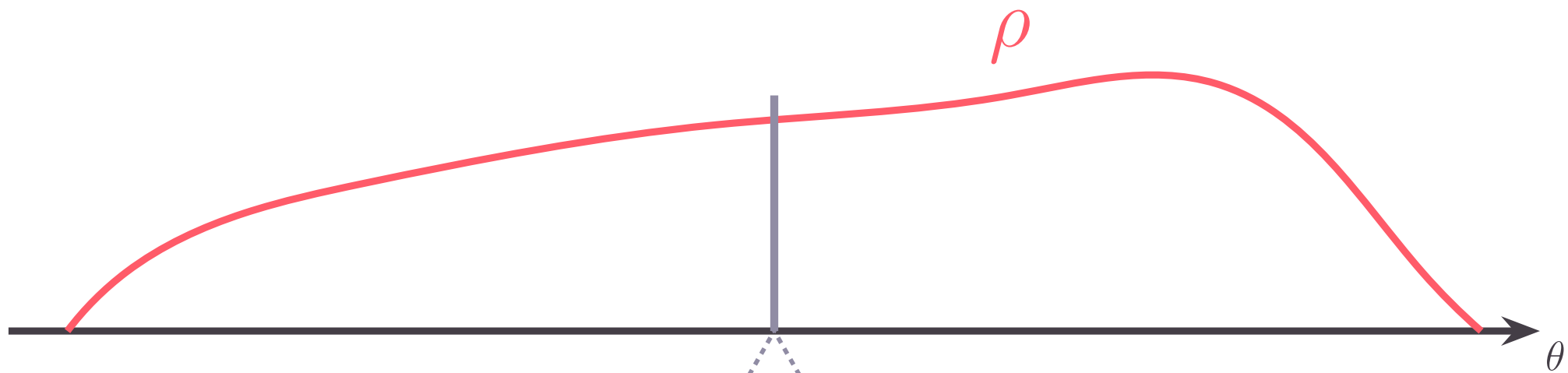


$L\rho(\theta)\delta\theta$: # rapidities $\in [\theta, \theta + \delta\theta]$

$L\rho_s(\theta)\delta\theta = \delta I$: # states $\in [\theta, \theta + \delta\theta[$

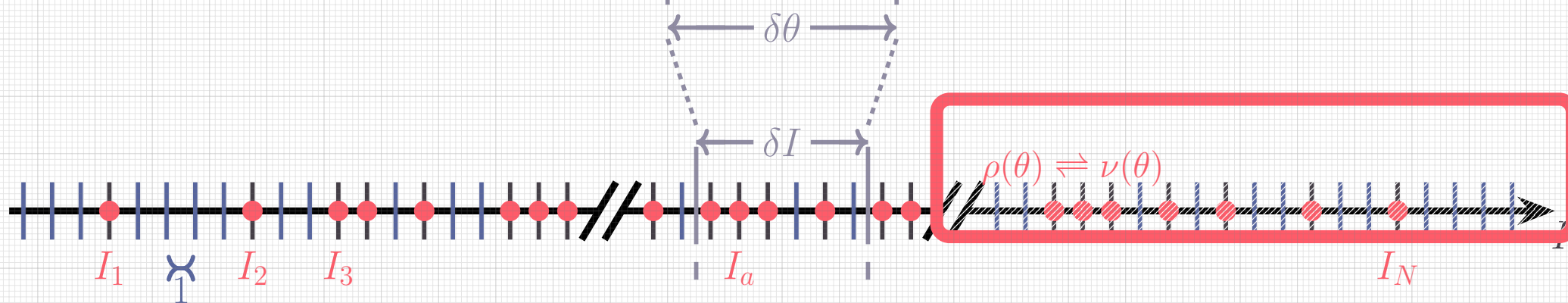
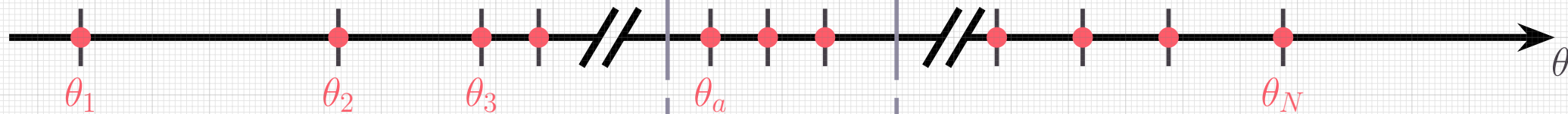
$\nu(\theta) = \frac{\rho(\theta)}{\rho_s(\theta)}$: Occupation factor

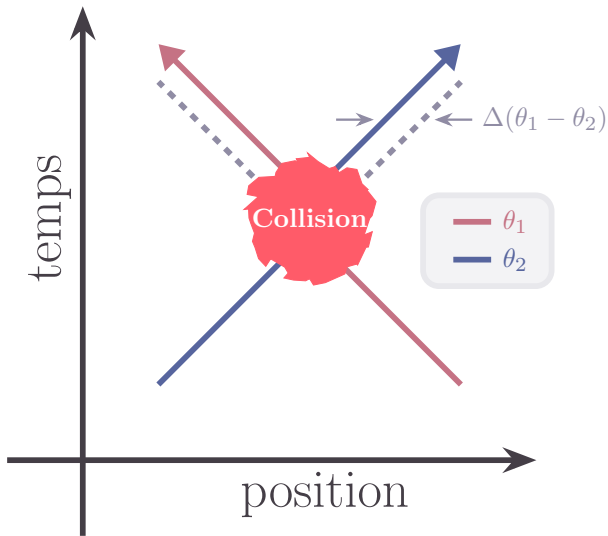
$\rho(\theta) \rightleftharpoons \nu(\theta)$

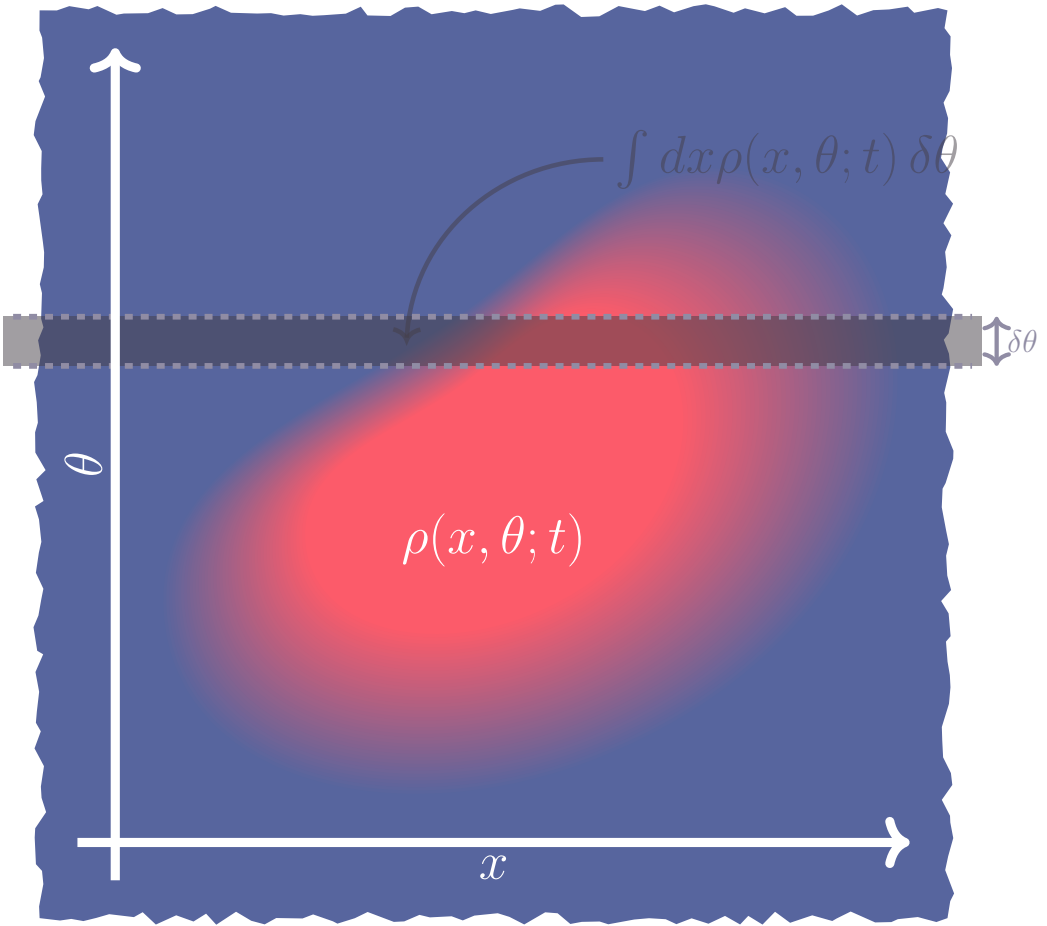


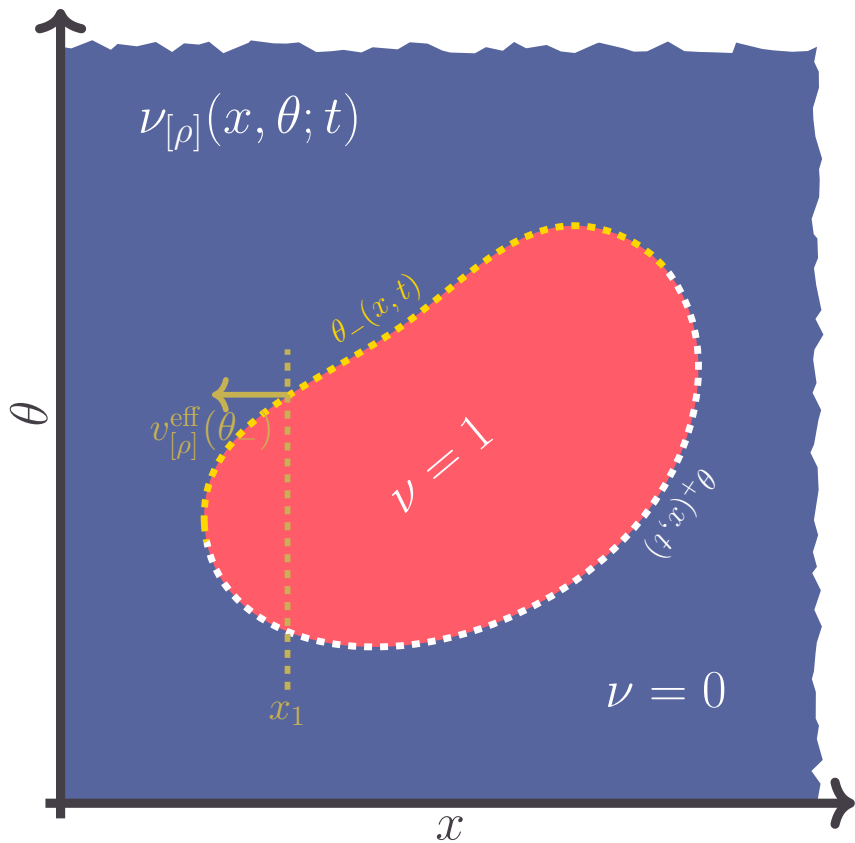


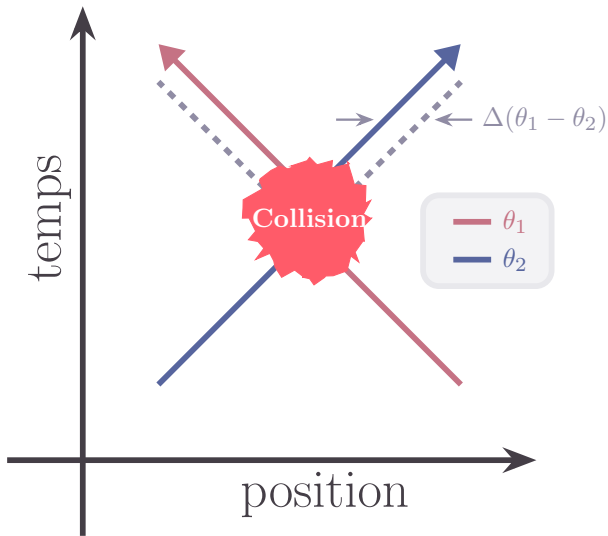
$$\nu(\theta) = \frac{\rho(\theta)}{\rho_s(\theta)} : \text{Occupation factor}$$



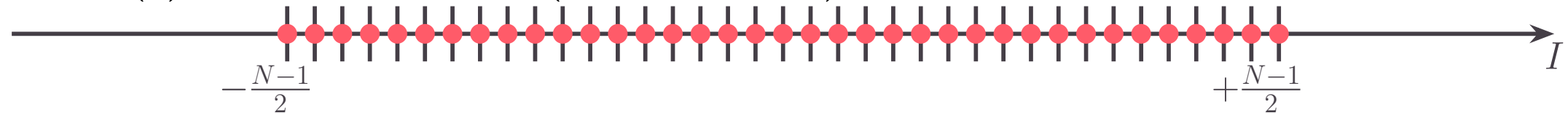




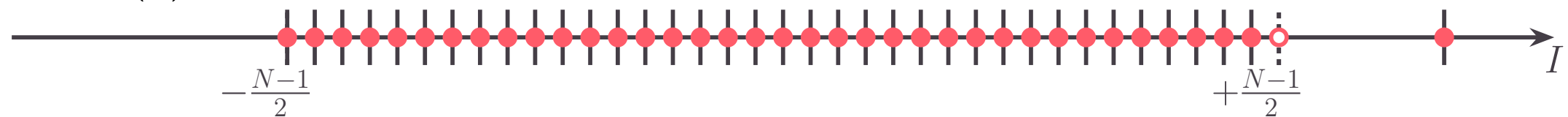




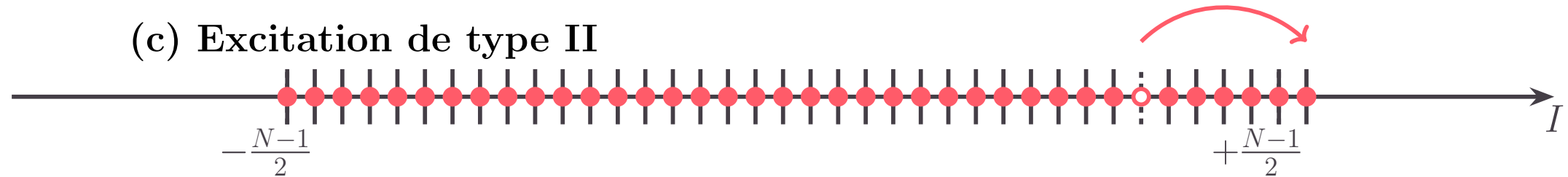
(a) État fondamental (Mer de Fermi)



(b) Excitation de type I



(c) Excitation de type II



$\bullet$: quasi-particule
 $\circ$: troue

ρ : distribution de rapidité
 ρ_h : densité de troue

$\rho_s = \rho + \rho_h$: densité d'état

